

E-Commerce Customer Retention & Cohort Analysis

Portfolio Project Proposal | Shopee/Lazada-style E-Commerce Marketplace | Ho Chi Minh City / Vietnam
Context

Project purpose: Build a job-ready data analytics portfolio project that demonstrates the ability to use SQL, Python, customer analytics, visualization, and business reasoning to diagnose retention problems and recommend actions for an e-commerce marketplace.

1. Executive Summary

This project will analyze customer purchasing behavior for a fictional Vietnam-focused e-commerce marketplace. The analysis will focus on customer acquisition, repeat purchasing, retention, cohort behavior, customer value, and churn risk. The final deliverable will resemble the type of analysis a junior Data Analyst could provide to growth, CRM, product, or commercial teams at an e-commerce company.

Rather than producing a collection of unrelated charts, the project will follow a business narrative: identify where retention weakens, determine which customer segments and acquisition patterns are associated with stronger repeat behavior, and translate the findings into practical recommendations.

2. Business Problem

Scenario. A Shopee/Lazada-style marketplace has continued to acquire new customers, but management is concerned that order growth depends too heavily on new-user acquisition and promotions. The company wants to understand whether customers are returning after their first purchase, which cohorts retain best, which customer groups generate the most value, and where retention initiatives should be focused.

Primary business question: How can the marketplace improve repeat purchasing and customer retention while focusing resources on the customer segments with the greatest long-term value?

3. Project Objectives

Objective	What the analysis should determine
Customer growth	How the active customer base changes over time and how much growth comes from new versus returning customers.
Retention	What percentage of customers return after their first purchase and how retention changes by cohort.
Customer value	Which customers and segments contribute the greatest revenue, order frequency, and estimated lifetime value.
Behavioral segmentation	How customers differ by recency, frequency, monetary value, category preference, discount usage, and geography.
Churn risk	Which previously active customers show signs of disengagement and could be targeted for reactivation.
Business action	Which customer groups should receive retention, loyalty, cross-sell, or reactivation initiatives.

4. Key Analytical Questions

1. How are monthly orders, revenue, active customers, and average order value changing?
2. What share of monthly customers are new versus returning?
3. What are the 30-, 60-, 90-, and 180-day repeat purchase rates?
4. How does retention differ across monthly acquisition cohorts?
5. Which first-purchase categories are associated with stronger subsequent retention?
6. Do customers acquired with large discounts return at the same rate as customers acquired with smaller discounts?
7. Which geographic areas or customer groups show unusually high or low retention?
8. Which RFM segments contain the highest-value loyal customers, promising new customers, and customers at risk?
9. How concentrated is revenue among high-value customers?
10. Which actionable interventions are supported by the data?

5. Proposed Dataset & Data Model

The project may use a public e-commerce dataset, a carefully constructed synthetic dataset, or a hybrid approach. If synthetic data is used, the repository and final report will clearly label it as simulated. The data should be large enough to support meaningful SQL analysis and multiple customer cohorts.

Field	Example purpose
customer_id	Unique customer identifier
order_id / order_item_id	Order and line-item identifiers
order_date	Time-series and cohort assignment
product / category	Product preference and category retention
quantity / price	Order economics and revenue
discount	Promotion dependency analysis
payment_method	Customer/payment behavior
city / district	Geographic segmentation
delivery_time	Optional service-experience variable
order_status	Completed, cancelled, returned, etc.

Recommended logical model

FactOrders / FactOrderItems linked to **DimCustomer**, **DimProduct**, **DimDate**, and optionally **DimGeography**. A star-schema-style structure will make the project more representative of professional analytics work than a single flat CSV.

6. Tools & Technical Scope

Tool	Role in project
SQL (PostgreSQL or similar)	Primary analytical querying: joins, CTEs, aggregations, CASE logic, date functions, subqueries, and window functions.

Tool	Role in project
Python / Pandas	Data cleaning, validation, deeper customer analysis, cohort matrices, RFM calculations, and reproducible analysis.
Matplotlib / Plotly	Exploratory and explanatory visualizations, including retention heatmaps and segment comparisons.
Power BI (optional but recommended)	Executive-facing summary dashboard to complement the deeper Python/SQL analysis.
Git / GitHub	Version control, documentation, reproducibility, and portfolio presentation.

7. Core Metrics

Metric group	Measures
Growth	Revenue, orders, active customers, new customers, returning customers, average order value
Retention	Repeat purchase rate, cohort retention by month, time to second order, retained customer rate
Value	Revenue per customer, order frequency, estimated customer lifetime value, contribution by segment
RFM	Recency, frequency, monetary score and customer segment
Promotion	Discount rate, retention by discount band, revenue/retention trade-off
Churn	Days since last order, inactive customer count, at-risk segment size

8. Analysis Methodology

- 1. Data audit and cleaning** - Inspect missing values, duplicates, invalid dates, cancelled/returned orders, inconsistent categories, outliers, and customer identifiers. Document all assumptions.
- 2. SQL exploratory analysis** - Create reusable SQL queries for monthly KPIs, customer ordering behavior, category performance, new/returning customers, and customer-level features.
- 3. Cohort analysis** - Assign customers to first-purchase cohorts and calculate month-by-month retention. Visualize the cohort matrix as a heatmap.
- 4. Repeat-purchase analysis** - Measure time to second purchase, repeat rate, purchase frequency, and differences by category, geography, and discount behavior.
- 5. RFM segmentation** - Score customers on recency, frequency, and monetary value, then create business-friendly segments such as Champions, Loyal, Promising, At Risk, and Lost.
- 6. Customer value analysis** - Estimate customer value using historical revenue/frequency and compare high-value versus low-value behavior. Avoid presenting a simple heuristic as a precise predictive CLV model.
- 7. Insight synthesis** - Select the findings that materially answer the business problem and distinguish correlation from causal claims.
- 8. Recommendation design** - Translate findings into specific CRM, loyalty, cross-sell, promotion, and reactivation recommendations, with suggested KPIs for measuring impact.

9. SQL Skills to Demonstrate

The SQL portion should be substantial enough that a recruiter can inspect it independently. The repository should include readable, commented queries using: **JOINS, CTEs, GROUP BY, CASE WHEN, date arithmetic, subqueries, ROW_NUMBER/RANK, LAG/LEAD, SUM/AVG OVER, partitioned window functions**, and customer-level aggregations.

Example outputs: first order per customer, cohort month, months since acquisition, monthly active customers, new-versus-returning customer counts, cumulative customer revenue, order sequence, time between purchases, and retention tables.

10. Planned Visualizations / Dashboard

Visual	Purpose
Monthly KPI trend	Show revenue, orders, customers, and repeat behavior over time.
New vs returning customers	Show whether growth is driven by acquisition or retention.
Cohort retention heatmap	Reveal how retention changes by acquisition cohort and customer age.
Time-to-second-order distribution	Show how quickly repeat customers return.
RFM segment chart	Summarize customer composition and value by behavioral segment.
Retention by first category / discount band	Identify behavioral patterns associated with stronger or weaker retention.
Revenue concentration	Show the contribution of high-value customers or segments.
Executive summary dashboard	Present the few KPIs and findings that management should monitor.

11. Final Deliverables

- Clean, documented dataset or reproducible data-generation/preparation process.
- SQL scripts organized by business question.
- Python notebook(s) or scripts for cleaning, cohort analysis, RFM, and deeper analysis.
- A polished retention/cohort visualization set and, ideally, a compact Power BI executive dashboard.
- GitHub README explaining the business problem, dataset, methodology, findings, recommendations, and how to reproduce the analysis.
- A short final case-study report suitable for sharing with recruiters.
- Resume-ready project bullets that quantify dataset size, analyses performed, tools used, and business outcomes identified.

12. Proposed Repository Structure

```
ecommerce-retention-analysis/  
|  
|-- data/  
|   |-- raw/  
|   |-- processed/  
|-- sql/  
|   |-- 01_data_quality.sql  
|   |-- 02_kpi_analysis.sql
```

```

|   |-- 03_customer_behavior.sql
|   `-- 04_cohort_retention.sql
|-- notebooks/
|   |-- 01_cleaning_eda.ipynb
|   |-- 02_cohort_analysis.ipynb
|   `-- 03_rfm_customer_value.ipynb
|-- dashboard/
|-- images/
|-- src/
|-- README.md
|-- requirements.txt
`-- data_dictionary.md

```

13. Suggested Four-Week Execution Plan

Week	Focus	Output
1	Dataset selection, business framing, data dictionary, database setup, cleaning	Validated dataset + schema + documented assumptions
2	SQL analysis and KPI layer	Organized SQL scripts + initial findings
3	Cohort retention, RFM, customer value, Python visualizations	Core analytical results + charts
4	Dashboard, recommendations, README, case-study polish	Recruiter-ready GitHub project

14. Success Criteria

The project will be considered portfolio-ready when a reviewer can understand the business problem within one minute, inspect substantial SQL work, see a correct cohort-retention methodology, understand the customer segments, identify at least three non-trivial insights, and connect each major insight to a practical business recommendation. The analysis should be reproducible and should clearly distinguish real data, simulated data, assumptions, descriptive findings, and any modeled estimates.

15. Portfolio Positioning

Target role: Junior Data Analyst / Data Analyst / BI Analyst / Product or Growth Analyst.

Primary skills demonstrated: SQL, Python/Pandas, data cleaning, cohort analysis, customer segmentation, retention metrics, visualization, business communication, and optional Power BI.

Portfolio message: I can take transactional customer data, define meaningful business metrics, analyze retention and customer value, and turn the results into recommendations that a commercial team can act on.